

Graph theory: structure, surfaces, spectra

Celebrating Bojan Mohar's 70th birthday



24–27 September 2026

Shanghai Center for Mathematical Sciences

Fudan University · Shanghai, China

Quick guide

Graph theory: structure, surfaces, spectra

Dates	24–27 September 2026
Venue	Gu Lecture Hall, Shanghai Center for Mathematical Sciences (SCMS) Fudan University, Jiangwan Campus, near Gate 2
Address	No. 2005 Songhu Road, Yangpu District, Shanghai 200438 上海市杨浦区淞沪路2005号

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Conference website and updates

scmscomb.github.io/conference

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About the meeting

This conference celebrates the work of Bojan Mohar on the occasion of his 70th birthday. It brings together researchers in graph theory, combinatorics and related areas, with themes spanning graph structure, graph embeddings, colouring, spectral methods, geometry and topology.

The meeting takes place from 24 to 27 September 2026 at the Shanghai Center for Mathematical Sciences, Fudan University.



Bojan Mohar

Host institution

Shanghai Center for Mathematical Sciences (SCMS), Fudan University
复旦大学上海数学中心

Plenary speakers

Name	Affiliation
Jørgen Bang-Jensen	University of Southern Denmark
Pavol Hell	Simon Fraser University
Ken-ichi Kawarabayashi	National Institute of Informatics
Daniel Král'	Leipzig University
Tomaž Pisanski	University of Primorska
Xuding Zhu	Zhejiang Normal University

Organizers

Name	Affiliation
Peter Bradshaw	University of South Carolina
Zdenek Dvořák	Charles University
Krystal Guo	University of Amsterdam
Ararat Harutyunyan	Université Paris Dauphine–PSL
Ken-ichi Kawarabayashi	National Institute of Informatics
Hehui Wu	SCMS, Fudan University

Local organizers and contacts

SCMS Combinatorics Group

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Thursday, 24 September

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Mid-Autumn Festival Banquet

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Jing Pu Hui · Shanghai Tower

晶浦会 · 上海中心大厦店

4th floor, Shop 2, Shanghai Tower

501 Yincheng Middle Road, Pudong New Area, Shanghai

上海市浦东新区银城中路501号

上海中心大厦上海之品商场4楼2号

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Talk abstracts

Thursday, 24 September

24 September • 10:00–10:25

Achieving Steffen's Edge-Colouring Bound

Jessica McDonald

Auburn University

We discuss the structure of graphs achieving known upper bounds for edge-colouring. In particular we focus on an upper bound of Steffen, and are able to prove that, under some mild assumptions, all critical graphs achieving the bound are so-called odd ring graphs. This is joint work with G. Chen, A. Fiujlaali, and A. Johnson-Yu.

24 September • 10:25–10:50

Diameter and Length of Metric Graphs

Sergio Cabello

University of Ljubljana

A metric graph is a metric space obtained from a finite, edge-weighted graph where the continuum of points along the interior of each edge is taken into consideration, and each edge is locally isometric to an interval whose length is the edge-weight. The diameter of a metric graph G is the maximum distance between all pairs of points of G . We show that the total length of a metric graph G with $l(G)$ leaves, cyclomatic number $\text{cyc}(G)$, and diameter $\text{diam}(G)$ is at most $(\text{cyc}(G) + \max\{1, l(G)/2\}) \text{diam}(G)$. Furthermore, we show that this bound is tight, and we characterize the metric graphs where equality holds.

24 September • 11:10–11:35

Separator Theorem for Minor-free Graphs in Linear Time

Tomáš Masařík

University of Warsaw

The planar separator theorem by Lipton and Tarjan [FOCS '77, SIAM Journal on Applied Mathematics '79] states that any planar graph with n vertices has a balanced separator of size $O(\sqrt{n})$ that can be found in linear time. This landmark result kicked off decades of research on designing linear or nearly linear-time algorithms on planar graphs. In an attempt to generalize Lipton-Tarjan's theorem to nonplanar graphs, Alon, Seymour, and Thomas [STOC '90, Journal of the AMS '90] showed that any minor-free graph admits a balanced separator of size $O(\sqrt{n})$ that can be found in $O(n^{3/2})$ time. The superlinear running time in their separator theorem is a key bottleneck for generalizing algorithmic results from planar to minor-free graphs. Despite extensive research for more than two decades, finding a balanced separator of size $O(\sqrt{n})$ in (linear) $O(n)$ time for minor-free graphs remains a major open problem. Known algorithms either give a separator of size much larger than $O(\sqrt{n})$ or have superlinear running time, or both. In this talk, we answer the open problem affirmatively. Our algorithm is very simple: it runs a vertex-weighted variant of breadth-first search (BFS) a constant number of times on the input graph. Our key technical contribution is a weighting scheme on the vertices to guide the search for a balanced separator, offering a new connection between the size of a balanced separator and the existence of a clique-minor model. We believe that our weighting scheme may be of independent interest.

This is a joint work with Édouard Bonnet, Tuukka Korhonen, Hung Le, and Jason Li.

24 September • 11:35–12:00

An Almost Quadratic Bound for the Minimal Excluded Minors for a Surface

Sarah Houdaigoui

National Institute of Informatics / SOKENDAI

As part of the Graph Minors Project, Robertson and Seymour showed in 1990 that the class of graphs embeddable in a given surface can be characterized by a finite set of minimal excluded minors. However, the proof is purely existential and therefore provides no explicit information about these excluded minors. In 1993, Seymour established the first general upper bound on the order of such minimal excluded minors. This bound is double-exponential in the Euler genus of the surface. Recently, we improved this result by deriving a quasipolynomial upper bound. Despite this advance, the gap between this bound and the best-known lower bound, which is linear, remained substantial. In this talk, I will present recent progress establishing that the bound can in fact be made almost quadratic in the Euler genus.

This is joint work with Ken-ichi Kawarabayashi.

24 September • 13:30–14:20

Results and open problems on acyclic dichromatic numbers of digraphs

Jørgen Bang-Jensen

Department of Mathematics and Computer Science, University of Southern Denmark, Odense, Denmark; School of Mathematics, Shandong University, Jinan, China

The **dichromatic number** $\vec{\chi}(D)$ of a digraph $D = (V, A)$ is the minimum number of sets in a partition $V_1, \dots, V_k$ of V into k subsets so that the induced subdigraph $D[V_i]$ is acyclic for each $i \in [k]$. This is a generalization of the chromatic number for undirected graphs as a graph has chromatic number at most k if and only if the complete biorientation G of G (replace each edge by a directed 2-cycle) has dichromatic number at most k . In this talk we introduce the **acyclic dichromatic number** $\vec{\chi}_a(D)$ of a digraph D as the minimum number of sets in a partition $V_1, \dots, V_k$ of V into k subsets so that the induced subdigraph $D[V_i]$ is acyclic for each $i \in [k]$ and each of the bipartite induced subdigraphs $D[V_i, V_j]$ is acyclic for each $1 \leq i < j \leq k$. We discuss a number of results which display the difference between the dichromatic number and the acyclic dichromatic number, in particular, there are digraphs D with arbitrarily large $\vec{\chi}_a(D) - \vec{\chi}(D)$, even among tournaments with dichromatic number 2 and bipartite tournaments (where the dichromatic number is always 2). We prove several complexity results, including that deciding whether $\vec{\chi}_a(D) \leq 2$ is NP-complete already for bipartite digraphs, while it is polynomial for tournaments (contrary to the case for dichromatic number). We also generalize the concept of **heroes** of a tournament to so-called **champions** of tournaments.

Based on joint work with Lucas-Picasarri-Arrieta and Anders Yeo.

24 September • 14:55–15:20

The Four Color Theorem with Linearly Many Reducible Configurations and Near-Linear Time Coloring

Atsuyuki Miyashita

The University of Tokyo

We give a quick overview of our recent result giving a near-linear-time algorithm for 4-coloring planar graphs, with particular emphasis on the additional ingredients needed beyond the previous quadratic-time algorithm of Robertson, Sanders, Seymour, and Thomas from 1996. Our key lemma states that every planar triangulation contains linearly many pairwise non-touching D-reducible configurations or pairwise non-crossing obstructing cycles. We explain the ideas behind the proof of this lemma and show how it enables us to construct a $O(n \log n)$ time 4-coloring algorithm.

24 September • 15:20–15:45

Three-edge-coloring apex cubic graphs

Yuta Inoue

The University of Tokyo

A graph G is apex if G has a vertex v such that $G - v$ is planar. We prove that every 2-connected apex cubic graph is three-edge-colorable. This result gives the final piece of the proof for the well-known Tutte's three-edge-coloring conjecture from 1966. The proof, as well as the result, generalizes that of the Four Color Theorem, which requires computer checks.

24 September • 16:30–16:55

Longest cycles in highly connected graphs

Jie Ma

Tsinghua University / University of Science and Technology of China

We will consider several problems on longest cycles in highly connected graph. We establish a higher-connectivity extension of Dirac's theorem and then use it to derive two applications: a conjecture of Bondy and a conjecture of Smith on longest cycles.

24 September • 16:55–17:20

Projection and Fiberning in Additive Combinatorics

Yifan Jing

The Ohio State University

I will present how to apply a spillover argument to control the combinatorial doubling of sets in fibers and under projections. A few earlier applications will be mentioned, including a nonabelian Brunn–Minkowski theory developed with Chieu-Minh Tran and Ruixiang Zhang. I will also mention a few new applications, including a solution of Ruzsa's conjecture, a solution of a conjecture of Christ and Iliopoulou, and a refinement of a theorem by Breuillard, Green, and Tao.

Friday, 25 September

25 September • 09:00–09:50

Ordering Characterizations of Graphs

Pavol Hell

Simon Fraser University

I will survey recent developments on ordering characterizations, many partially motivated by my joint paper with Bojan Mohar and Arash Rafiey. Work of, and work with, Barsukov, Bodirsky, Bok, Feder, Feller, Feuillolay, Guzman-Pro, Hadek, Habib, Hernandez-Cruz, Huang, Jedličková, Kun, Lin, Madelaine, Nešetřil, Paul, Perinti, Protopapas, Stewart, Vardi, and others will be discussed.

25 September • 10:00–10:25

Motley Mathematical Morsels of Mohar

Peter Mark Kayll

University of Montana

To his colleagues, Bojan Mohar is known for his open-door approach to collaboration. Natural consequences: lots of coauthors, a wide impact across the mathematical landscape, and many math friends. This talk will review a few of Professor Mohar's theorems buried deep in the modest vault of joint articles written with the speaker.

25 September • 10:25–10:50

The Proper Conflict-Free Coloring

Riste Škrekovski

University of Primorska

A proper vertex coloring φ of a graph G is said to be conflict-free if, for each non-isolated vertex $x \in V(G)$, there exists a color c such that $\varphi^{-1}(c) \cap N(x)$ has size one. In other words, in the neighborhood of every non-isolated vertex, some color appears exactly once.

In this talk, I will present some results and problems concerning the degree-choosability of conflict-free colorings, obtained jointly with R. Xu (China) and M. Kashima (Japan).

25 September • 11:10–11:35

Nice Subgraphs in Cubic Graphs, 6-Cuts, and the Berge Conjecture

Martin Škoviera

Comenius University Bratislava

Joint work with Edita Máčajová and Roman Nedela.

A subgraph of a cubic graph is nice if its complementary subgraph admits a perfect matching. In this paper we investigate nice subgraphs separated from the rest of the graph by edge cuts of size six or less. As an application we show that every bridgeless cubic graph with colouring defect at most 4 can have its edges covered with at most five perfect matchings, and hence satisfies the celebrated Berge conjecture. This improves an earlier result of Steffen (J. Graph Theory, 2015), which makes use of a classical result of Seymour (Proc. London Math Soc., 1979) and requires the graph not to have essential 3-edge-cuts.

25 September • 11:35–12:00

Well-Quasi-Ordering Eulerian Digraphs

Stephan Kreutzer

Technische Universität Berlin

A well-known result by Robertson and Seymour establishes that the class of undirected graphs is well-quasi-ordered by the minor and also by the weak immersion relation. For directed graphs the situation is much more dire as many natural digraph relations have been shown not to be a well-quasi-order on the class of all digraphs. Eulerian digraphs form an important subclass with often much better properties than general digraphs and we conjecture that Eulerian digraphs are well-quasi-ordered under the immersion relation. In joint work with Dario Cavallaro and Ken-ichi Kawarabayashi we have recently shown that classes of Eulerian digraphs embeddable on a fixed surface are well-quasi-ordered under the immersion relation. In this talk I will present some of the main ideas of this result and report on further progress in this ongoing project. This is joint work with Dario Cavallaro and Ken-ichi Kawarabayashi.

Saturday, 26 September

26 September • 09:00–09:50

Alon-Tarsi number of line graphs

Xuding Zhu

Zhejiang Normal University

Let G be a multigraph and $L(G)$ be the line graph of G . Let $P_{L(G)}(x_e : e \in E(G)) = \prod_{ee' \in E(L(G)), e < e'} (x_e - x_{e'})$ be the graph polynomial of $L(G)$ (where $<$ is an arbitrary linear order of $E(G)$). The Alon-Tarsi number of $L(G)$, denoted by $AT'(G)$, is the minimum integer k such that $P_{L(G)}$ has a non-vanishing monomial $\prod_{e \in E(G)} x_e^{t(e)}$ with $t(e) < k$ for all $e \in E(G)$. It follows from Combinatorial Nullstellensatz that for any graph G , $\chi'_t(G) \leq AT'(G)$, where $\chi'_t(G)$ is the list chromatic index of G . For a simple graph G and a positive integer r , let $G^{(r)}$ be the multigraph obtained from G by replacing each edge with r parallel edges. We prove that (1) if G is a simple series-parallel graph with $\Delta(G) \geq 3$, then $AT'(G^{(r)}) = \Delta(G^{(r)}) = r \cdot \Delta(G)$ (joint work with Deng). (2) For any multicircuit G (i.e., a multigraph whose underlying simple graph is a cycle), $AT'(G) = \chi'_t(G) = \chi'(G) = \max\{\Delta(G), \lceil \Gamma(G) \rceil\}$ (joint work with Chen, Kozik, Long and Zhu). (3) If n, r are positive integers, then $rn \leq AT'(K_{n,n}^{(r)}) \leq rn + 1 = \Delta(K_{n,n}^{(r)}) + 1$, and if n is odd, then the upper bound is the correct value (joint work with Chen, Kozik, Long and Zhu, and the $r = 1$ case was proved by Janssen). Alon conjecture that if n is even, $AT'(K_{n,n}) = n$ and the conjecture remains open. Some other results on this parameter will also be surveyed and some questions will be asked.

26 September • 10:00–10:25

Dominating Models

David R. Wood

Monash University

A dominating K_t -model in a graph G is a sequence $(T_1, \dots, T_t)$ of pairwise vertex-disjoint connected subgraphs of G such that, whenever $1 \leq i < j \leq t$, every vertex in T_j has a neighbour in T_i . Replacing “every vertex in T_j ” by “some vertex in T_j ” retrieves the standard definition of a K_t -model, which is equivalent to a K_t -minor in G .

This concept was introduced by Illingworth and Wood [Journal of Graph Theory 110(4): 448–456, 2025]. This talk surveys results on dominating models, including the recent result of Girão, Illingworth, Mohar, Norin, Steiner, Tamitegama, Tan, Wood, and Yip [arXiv:2605.10112], which shows that every graph with no dominating K_5 -model is 4-colourable. This generalizes and is significantly stronger than the Four Colour Theorem for planar graphs or for graphs with no K_5 -minor. It also makes progress towards Hajós’ conjecture on K_5 -subdivisions in 5-chromatic graphs.

26 September • 10:25–10:50

Discrete Steklov Eigenvalues of Self-Similar Networks

Haiyan Chen

Jimei University

Over a century ago, Steklov introduced the Steklov problem during the study of liquid sloshing. Since then, the eigenvalues of the Steklov operator have been intensively studied for Riemannian manifolds with boundary. Recently, Hua–Huang–Wang and Hassannezhad–Miclo independently extended the Steklov problem to discrete spaces. Numerous studies have since investigated the properties of Steklov eigenvalues on graphs, such as upper bounds for the first Steklov eigenvalue on trees, the monotonicity of Steklov eigenvalues, and lower bounds for Steklov eigenvalues. In this talk, we first show that Steklov eigenvalues on a graph are closely related to the effective resistances among its boundary vertices. We then give the explicit Steklov eigenvalues of several self-similar networks with special boundaries, including the Sierpiński gasket and the pentagasket network.

26 September • 11:10–11:35

Dichoosability of Graphs with High Choosability

Gil Puig i Surroca

Université Paris Dauphine–PSL

A conjecture of Erdős and Neumann-Lara from 1979 says that if every orientation of a graph G can be partitioned into k acyclic subgraphs, then the chromatic number of G is at most $f(k)$, where f is a function that is independent of G . We will discuss the list version of this conjecture and explain how it can be approached using ideas of Alon about the choosability of graphs with large minimum degree. This is joint work with Ararat Harutyunyan and Lucas Picasarri-Arrieta.

26 September • 11:35–12:00

Characteristic Polynomial of Uniform Hypertrees

Honghai Li

Jiangxi Normal University

We express the characteristic polynomial of any uniform hypertree in terms of matching polynomials of its connected subtrees. Some classical results in the literature are generalized to uniform hypergraphs, and a conjecture by Clark and Cooper is resolved completely.

26 September • 13:30–14:20

On Polycirculant Graphs

Tomaž Pisanski

University of Primorska

An automorphism a of a graph G , viewed as a permutation of the vertex set, is semiregular if it can be written as a product of k disjoint cycles of equal length, say $m > 1$. A graph G paired with its semiregular automorphism a is called a polycirculant structure (G, a) , and any graph that admits a polycirculant structure is called a polycirculant graph.

Polycirculant structures can be approached from different perspectives, making them the subject of intensive research not only in algebraic graph theory and permutation group theory, but also more broadly in topological graph theory, incidence geometry, discrete geometry, theoretical computer science, algebraic topology, and related fields. In this talk, we give an overview of some useful applications of the theory of polycirculants to various areas of discrete mathematics.

The driving force behind the research on polycirculants has been the almost half-a-century-old polycirculant conjecture, which asks whether every vertex-transitive graph has a semiregular automorphism. The conjecture was recently disproved by S. D. Freedman and M. Lee, who constructed infinitely many vertex-transitive graphs that are not polycirculants.

A polycirculant family of graphs consists of graphs that share the same quotient graph under a covering projection. We present some results on polycirculant families of graphs. Since polycirculants admit concise descriptions via cyclic voltage graphs, the space needed for storing large datasets of such graphs may be reduced considerably.

The presentation draws on my recent and ongoing work with many coauthors. This research is supported in part by the Slovenian Research Agency through research program P1-0294 and projects J1-4351 and J5-4596.

26 September • 14:30–14:55

Four Decades of Gratitude to Bojan Mohar

Drago Bokal

University of Maribor

Bojan trained an eye, and this is what it learned to see. A Takao Tanabe retrospective in Vancouver, read as one arc — the chaos of cases climbing into veristic clarity — and every proof I have written since climbs that same slope. He lent me a lens for crossing numbers: read the structure, not the cases. Through it came the zip product, crossing-critical graphs of every rational average degree in $(3, 6)$, the bounded tile catalogue that characterises the 2-crossing-critical graphs, and the threshold where degree stays bounded to twelve crossings and breaks free at thirteen. Then the lens turned to interpretations — crossings read as a resource, scarcity forcing rigidity and abundance yielding variety — and the arc leaves mathematics carrying three gifts, three values, the corners of one triangle. The will to gratitude. An impossibility theorem — Bowles's social trilemma — forbids any neutral rule that at once keeps a society ordered, voluntary, and growing; one leg must always give. The theorem opens the choice as a responsibility — and what meets it is gratitude: to whoever upholds the order, or yields the resources that let it grow, honouring those who give of their own for the rest. Order. Model a maturing population as subjects curving through an ordinal panel data: laminar when all advance by one order and none overtakes, turbulent when they overtake — each overtaking a crossing. Ranking becomes crossing-number minimization; order runs laminar, a disrupted world turbulent, and which hidden surface smooths the crossings out is open. Growth. A lens is borrowed only to be handed on: Tarča, a small game built by a high-school student where elementary algebra meets planar graphs, draws the young in, and workshops where undergraduates found their first theorems keep the path open. A lineage grows only by handing the young a game worth winning. Gratitude, order, growth — no neutral rule holds all three at once, so the will to gratitude must, taken up as a shared responsibility. The arc that carries them is itself gratitude, generational: from curiosity in the chaos of youth, to the order I first found in coding, the climb that brought me to Tomo, my mathematical grandfather, then to Bojan, and now brings my own students to me — owed to people, not their works. Purpose divides into pleasure, power, and meaning; gratitude unifies. Not the blossom admired, but the fruit returned. Thank you, Bojan.

26 September • 14:55–15:20

Toroidal graphs in orientable surfaces

Gašper Fijavž

University of Ljubljana

The embedding of a graph G in a nonspherical surface Σ is k -representative if no union of $< k$ faces of the embedding of G contains an essential curve. Using the excluded minor theorem for K_5 it is easy to see that every graph G admitting a 3-representative embedding in a nonspherical surface Σ contains a K_5 minor. Krakovski and Mohar have established that K_6 is a minor in every embedding of representativity ≥ 7 .

Working with K_7 we hit two problems. On one hand K_7 does not embed in the projective plane or the Klein bottle, on the other working with nonorientable surfaces of higher genera introduces additional topological difficulties. Nevertheless, using a result of Brunet, Mohar and Richter on homotopic cycles in highly representative graphs we were able to show that every graph admitting a 19-representative embedding in an arbitrary *orientable* surface Σ contains a K_7 -minor. In this talk we extend the result to arbitrary toroidal graphs: we show that for every toroidal graph T there exists a constant c_T so that for every graph G and for every orientable surface Σ if G admits an c_T -representative embedding in Σ then T is a minor of G .

26 September • 15:20–15:45

Drawing a Disconnected Graph on the Torus with Help from Minkowski

Robert Šámal

Charles University

Joint work with Bojan Mohar. We study drawings of graphs on the torus in which edges are allowed to cross. Let $cr_i(G)$ denote the minimum number of crossings in a drawing of G on the orientable surface of genus i . For a disjoint union $H = G \dot{\cup} G'$, drawing one graph in a disc and the other on the torus gives the obvious upper bound $cr_1(H) \leq \min\{cr_0(G) + cr_1(G'), cr_1(G) + cr_0(G')\}$. We prove that equality always holds. Equivalently, there is an optimal drawing of H on the torus in which the two graphs do not cross each other; in fact, one of them can be drawn inside a disc. The proof translates the topological problem into a two-dimensional lattice inequality. A decomposition theorem for Eulerian graphs represents the essential closed curves of the two drawings by multisets $X, X' \subseteq \mathbb{Z}^2$. Curves represented by u and v must meet at least $|\det(u, v)|$ times. Thus the crossings between the two graphs are bounded below by $L(X, X') = \sum_{u \in X} \sum_{v \in X'} |\det(u, v)|$. On the other hand, cutting the torus along a lattice basis and reconnecting the resulting arcs in a disc gives a planar drawing whose cost is a quantity $M(X)$; changing the basis corresponds exactly to applying a unimodular integer matrix. The key estimate is $L(X, X') \geq M(X)M(X') \geq \min\{M(X), M(X')\}$. We prove it by associating a centrally symmetric zonotope to each multiset. The argument combines Minkowski's inequality for mixed areas, Minkowski's theorem on successive minima, and Mahler's inequality for polar bodies.

26 September • 16:30–16:55

Cops and Robbers on Token Graphs

Seyyed Aliasghar Hosseini

Simon Fraser University

Given a graph and a positive integer k , the k -token graph has as vertices the k -subsets of V , with two configurations adjacent whenever one can be obtained from the other by sliding a single token along an edge of G . Token graphs are often much larger and more complicated than the original graph, making even basic graph parameters difficult to determine.

In this talk, we study the classical game of Cops and Robbers on token graphs. We introduce an equivalent interpretation in which the game is played directly on G : the robber controls a team of tokens, while each cop corresponds to a team of tokens. This formulation allows pursuit strategies on $\tau_k(G)$ to be understood in terms of the structure of the original graph.

Using this viewpoint, we determine the exact cop number of token graphs of paths and stars, and obtain corresponding results for subdivided stars. We also discuss results for cycles and some open questions on the cop number of token graphs.

26 September • 16:55–17:20

The Uniform Turán Density of The Tetrahedron

Xichao Shu

Leipzig University

Turán's Tetrahedron Problem, posed in 1941, asks for the Turán density of the complete 3-uniform hypergraph $K_4^{(3)}$, and remains one of the central open problems in extremal combinatorics. In the 1980s, Erdős and Sós proposed a uniformly distributed version of the problem, asking for the uniform Turán density of $K_4^{(3)}$. While the corresponding problem for the broken tetrahedron $K_4^{(3)-}$ was settled about a decade ago, the tetrahedron case remained open. We solve this problem by showing that the uniform Turán density of $K_4^{(3)}$ is $1/2$, confirming that a construction of Rödl from 1986 is optimal. This is joint work with Bartłomiej Kielak, Daniel Král', Ander Lamaison, Hong Liu, and Zhuo Wu.

Sunday, 27 September

27 September • 09:00–09:50

Structural sparsity in algorithm design and optimization

Daniel Král'

Leipzig University and MPI-MiS

Depth and width parameters of graphs, e.g., tree-width, path-width and tree-depth, play a central role in modern algorithmic and structural graph theory. These notions are of fundamental importance in the theory of graph minors, fixed parameter complexity and combinatorial sparsity. In this talk, we survey some of many structural and algorithmic results related to combinatorial sparsity. We will highlight a link between combinatorial sparsity of matroids and Graver bases, which are essential in the design of iterative integer programming algorithms. As an application of this link, we will present matroid based algorithms for detecting a hidden Dantzig-Wolfe-like structure in integer programs and uncovering it when present, which enables solving such instances efficiently.

27 September • 10:00–10:25

A few words about maps

Patrice Ossona de Mendez

CNRS

We survey some properties, encoding, and bijections involving combinatorial maps, double occurrence words, and chord diagrams. We particularly study quasi-trees from a purely combinatorial point of view and derive a topological representation of maps with a given spanning quasi-tree using two fundamental polygons, which extends the representation of planar maps based on the equivalence with bipartite circle graphs. Then, we focus on Depth-First Search trees and their connection with a poset we define on the spanning quasi-trees of a map and discuss applications to enumeration problems.

(Joint work with Robert Cori and Pierre Rosenstiehl)

27 September • 10:25–10:50

A generalisation of Ramsey's theorem and its applications to degree-boundedness

Lucas Picasarri-Arrieta

National Institute of Informatics

We prove a new generalisation of Ramsey's theorem by showing that every 2-edge-coloured graph with sufficiently large minimum degree contains a monochromatic induced subgraph whose minimum degree remains large.

From this, we also derive that every orientation of a graph with large minimum degree contains either a large transitive tournament or an induced antidirected digraph whose minimum degree is still large.

As a consequence, we obtain two general tools showing that certain extensions of degree-bounded graph classes preserve degree-boundedness. A hereditary class $\mathcal{G}$ is degree-bounded if, for every integer s , there exists $d = d(s)$ such that every graph $G \in \mathcal{G}$ either contains $K_{s,s}$ or has minimum degree at most d . With these tools, we obtain for instance that odd-signable graphs and Burling graphs are degree-bounded. We also characterise exactly the oriented graphs F such that the graphs admitting an orientation without any induced copy of F are degree-bounded.

Joint work with Arnab Char and Ken-ichi Kawarabayashi.

27 September • 11:10–11:35

Seymour-tight Orientations

Gabriëlle Zwaneveld

University of Amsterdam

In this talk, we explore ‘almost counterexamples’ to Seymour’s second neighbourhood conjecture. In what we call *Seymour-tight orientations*, the size of the first neighbourhood of each vertex equals the size of its second neighbourhood. We give several examples and constructions. Specifically, we prove that the class of Seymour-tight orientations is closed under taking (generalized) lexicographic products.

Moreover, the lexicographic product of a putative counterexample to Seymour’s second neighbourhood conjecture and a Seymour-tight orientation is again a counterexample. Using lexicographic products, we show that if the conjecture is false, then there exist counterexamples that are close to regular tournaments, and moreover that any digraph occurs as an induced subgraph of a counterexample.

Lastly, using Kemperman’s theorem, we characterize the Seymour-tight orientations that are the Cayley digraph of an abelian group.

27 September • 11:35–12:00

Tilings of Hyperbolic Surfaces

Min Yan

The Hong Kong University of Science and Technology

We consider tilings of closed surfaces of constant curvature -1 and Euler characteristic $\chi < 0$ by congruent n -gons. For fixed χ and $n \geq 7$, the tiling problem is actually a finite problem. Two extremes of the problem are single-tile tilings and tilings with the maximal number of tiles. The study of the two extremes has very different characteristics, and we concentrate on single-tile or double-tile edge-to-edge tilings.

Combinatorially, single-tile tilings are the same as planar diagrams. We discuss two algorithms for obtaining all single-tile tilings over a fixed surface and list these tilings for surfaces of small genus. We discuss the relation between our results and earlier work by Zamorzaeva–Orleanschi and by Harer and Zagier.

Geometrically, we discuss the geometric realization of the combinatorial tilings, the universal covers of tilings, and isometries of the surfaces underlying the tilings.

We also briefly discuss two-tile tilings, where some new phenomena appear: the number of distinct edge lengths and the positive-angle-value condition.

27 September • 13:30–13:55

Turán problems in planar graphs

Ervin Győri

Alfréd Rényi Institute of Mathematics

Let $\text{ex}_p(n, T, H)$ denote the maximum number of copies of T in an n -vertex planar graph which does not contain H as a subgraph. When $T = K_2$, $\text{ex}_p(n, T, H)$ is the planar Turán number of H , denoted by $\text{ex}_p(n, H)$. The topic of extremal planar graphs was initiated by Dowden (2016) [1]. He obtained sharp upper bound for both $\text{ex}_p(n, C_4)$ and $\text{ex}_p(n, C_5)$. In [2], we gave a sharp upper bound $\text{ex}_p(n, C_6) \leq 5n/2 - 7$ for all $n \geq 18$.

We proved that for every integer $n \geq 6$, $\text{ex}_p(n, C_5, \emptyset)$ is $2n^2 - 10n + 12 + (1)$ (see [3]). And (see [4]), for every fixed $k \geq 3$, $\text{ex}_p(n, C_{2k}, \emptyset)$ is $n^k/k^k + o(n^k)$. In this lecture, we present more recent similar results related to cycles and paths.

Keywords: planar graph; extremal graph.

27 September · 13:55–14:20

The Unstable Dual of a 2-Cell Embedding

MacKenzie Carr

Toronto Metropolitan University

Given a graph G , its genus distribution is defined as the sequence $\{g_k(G)\}_{k \geq 0}$, where $g_k(G)$ is the number of 2-cell embeddings of G in the orientable surface of genus k . The Log-Concavity Genus Distribution (LCGD) Conjecture, first posed in 1989, claims that the genus distribution is a log-concave sequence. However, this was recently shown by Mohar to be false for 4-connected graphs. One reason that the LCGD Conjecture remained unsolved for so long is the difficulty of computing the terms of the genus distribution, particularly for large graphs. Using local rotations, we introduce a new way of describing the space of 2-cell embeddings and their mutual relationships for any fixed cubic graph. We introduce the unstable dual of an embedding of a cubic graph, a subgraph of the dual graph, and describe how the genus of the corresponding embedding can be recovered from properties of the unstable dual. Finally, we discuss how the unstable dual can be used to transform the calculation of the genus distribution of a cubic graph into a subgraph-counting problem.

Discussion notes

Venue and accommodation

SCMS • Conference venue

上海数学中心·会议地点

Gu Lecture Hall, Shanghai Center for Mathematical Sciences

Fudan University, Jiangwan Campus

Address No. 2005 Songhu Road, Yangpu District,

Shanghai 200438, China

上海市杨浦区淞沪路2005号

Near Gate 2 · 复旦大学江湾校区2号门附近

Fraser Place Wujiaochang Shanghai

上海五角场辉盛坊公寓酒店

Address No. 1258 Yinhang Road, Yangpu District,

Shanghai 200438, China

上海市杨浦区殷行路1258号

Nearest metro station: Xinjiangwancheng · 新江湾城站

SCMS Guest House

上海数学中心专家楼

Address Opposite Lane 27, Xuede Road,

Xinjiangwan City Subdistrict, Yangpu District, Shanghai, China

上海市杨浦区新江湾城街道学德路27弄对面

Beside Gate 3 · 复旦大学江湾校区3号门旁

From your accommodation to SCMS

Fraser Place: A bus will take participants to SCMS each morning. The venue is also approximately a 25-minute walk from the hotel.

SCMS Guest House: SCMS is within a 15-minute walk, via Gate 3.

Airport and railway connections

All routes below use Metro Line 10 towards **Jilong Road (基隆路)**. Choose your destination first; the final station and walking directions differ.

1 • SCMS

From	Metro route to Guofan Road
Pudong Airport (PVG)	Line 2 → East Nanjing Road; change to Line 10 → Guofan Road.
Hongqiao Airport (SHA) Terminals 1 and 2	Line 10 → Guofan Road (direct).
Hongqiao Railway Station	Line 10 → Guofan Road (direct).
Shanghai Railway Station	Line 4 (inner loop, via Baoshan Road) → Hailun Road; change to Line 10 → Guofan Road.

On foot: Leave Guofan Road Station (国帆路站) at **Exit 3**; enter the Jiangwan Campus through **Gate 2** and walk to SCMS (approximately 10 minutes).

2 • Fraser Place Wujiaochang Shanghai

From	Metro route to Xinjiangwancheng
Pudong Airport (PVG)	Line 2 → East Nanjing Road; change to Line 10 → Xinjiangwancheng.
Hongqiao Airport (SHA) Terminals 1 and 2	Line 10 → Xinjiangwancheng (direct).
Hongqiao Railway Station	Line 10 → Xinjiangwancheng (direct).
Shanghai Railway Station	Line 4 (inner loop, via Baoshan Road) → Hailun Road; change to Line 10 → Xinjiangwancheng.

On foot: From Xinjiangwancheng Station (新江湾城站), walk approximately **600 m** to Fraser Place at **1258 Yinhang Road (殷行路 1258 号)**.

3 • SCMS Guest House

From	Metro route to Xinjiangwancheng
Pudong Airport (PVG)	Line 2 → East Nanjing Road; change to Line 10 → Xinjiangwancheng.
Hongqiao Airport (SHA) Terminals 1 and 2	Line 10 → Xinjiangwancheng (direct).
Hongqiao Railway Station	Line 10 → Xinjiangwancheng (direct).
Shanghai Railway Station	Line 4 (inner loop, via Baoshan Road) → Hailun Road; change to Line 10 → Xinjiangwancheng.

On foot: Leave Xinjiangwancheng Station at **Exit 6 or 7** and walk towards **Gate 3**. The Guest House is beside the gate, opposite Lane 27, Xuede Road (学德路 27 弄对面).

Local map



Navigation

1 • Jiangwan Campus • SCMS

复旦大学江湾校区·上海数学中心

2005 Songhu Road
淞沪路2005号

Conference venue • Gate 2
会场·2号门附近



[Google Maps](#)

2 • U-Fun shopping centre

悠方购物中心

1660 Songhu Road
淞沪路1660号

Shopping and dining
购物与餐饮



[Google Maps](#)

3 • Fudan • Handan Campus

复旦大学邯郸校区

220 Handan Road
邯郸路220号

Campus visit
校园参观



[Google Maps](#)

4 • Daxue Road

大学路

Daxue Road, Yangpu
杨浦区大学路

Cafés, food and tea
咖啡、餐饮与茶饮



[Google Maps](#)

5 • Wujiaochang

五角场商圈

Yangpu District, Shanghai
上海市杨浦区五角场

Shopping and dining
购物与餐饮



[Google Maps](#)

6 • Jiangwanli • MEET678

江湾里 MEET678

678 Minfu Road
民府路678号

Shopping and dining
购物与餐饮



[Google Maps](#)

Meals and practical information

Meals at the meeting

Lunch is served at the dining hall. Meal vouchers are provided for breakfast and dinner at the dining hall.

Mid-Autumn Festival Banquet

Evening of Friday, 25 September

Jing Pu Hui · Shanghai Tower

晶浦会 · 上海中心大厦店

4th floor, Shop 2, Shanghai Tower

501 Yincheng Middle Road, Pudong New Area, Shanghai

上海市浦东新区银城中路501号

上海中心大厦上海之品商场4楼2号

Campus visits

Conference campus Jiangwan Campus · 2005 Songhu Road
复旦大学江湾校区 · 淞沪路2005号

Handan Campus Handan Campus · 220 Handan Road
复旦大学邯郸校区 · 邯郸路220号

For campus access arrangements, contact the local organizers on page 4.

Internet and payment

Eduroam is available at the conference venue, and Wi-Fi is available at the hotels.

Alipay and WeChat are widely used for local payments.

Food and shopping nearby

U-Fun · 悠方 Shopping and dining near the Jiangwan Campus.

Daxue Road · 大学路 Cafés, tea and casual dining.

Wujiaochang · 五角场 Shopping centres and restaurants.

Writing space